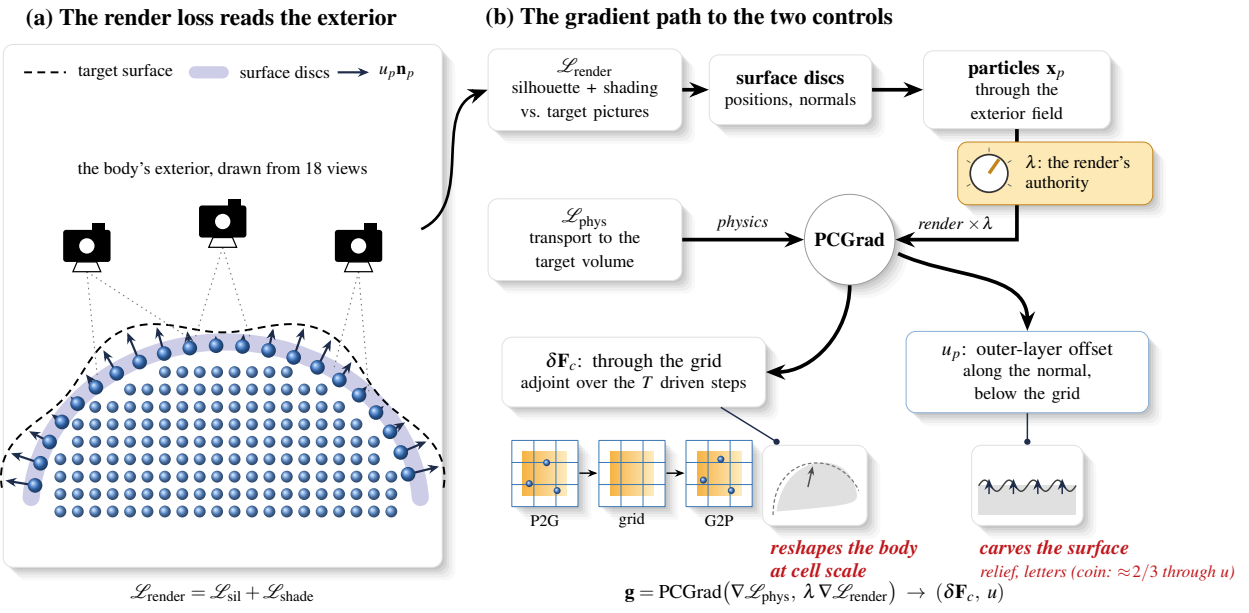

(a) The render loss reads the exterior
target surface
surface discs
$u_p \mathbf{n}_p$
the body's exterior, drawn from 18 views
$\mathscr{L}_{\text{render}} = \mathscr{L}_{\text{sil}} + \mathscr{L}_{\text{shade}}$
(b) The gradient path to the two controls
$\mathscr{L}_{\text{render}}$
silhouette + shading
vs. target pictures
surface discs
positions, normals
particles $\mathbf{x}_p$
through the
exterior field
$\lambda$: the render's
authority
$\mathscr{L}_{\text{phys}}$
transport to the
target volume
physics
PCGrad
render $\times \lambda$
$\delta \mathbf{F}_c$: through the grid
adjoint over the $T$ driven steps
$u_p$: outer-layer offset
along the normal,
below the grid
P2G
grid
G2P
reshapes the body
at cell scale
carves the surface
relief, letters (coin: $\approx 2/3$ through $u$)
$\mathbf{g} = \text{PCGrad}\big(\nabla \mathscr{L}_{\text{phys}},\ \lambda \nabla \mathscr{L}_{\text{render}}\big) \ \rightarrow \ (\delta \mathbf{F}_c,\ u)$